Hybrid Quantum-Classical Transfer Learning Model for Credit Risk Assessment with Reduced Quantum Overhead and Enhanced Minority Class Prediction

1st Parvathy Gopakumar
QuDAIS Lab
Indian Institute of Information Technology Kottayam
Kerala, India
parvathymlm.22phd11009@iiitkottayam.ac.in

2nd Rubell Marion Lincy G
QuDAIS Lab
Indian Institute of Information Technology Kottayam
Kerala, India
lincy@iiitkottayam.ac.in

3rd Salvatore Sinno
Enterprise Computing Solutions,Unisys
England, UK
salvatore.sinno@gb.unisys.com

4th Shruthi Thuravakkath
Enterprise Computing Solutions,Unisys
Bangalore, Karnataka, India
Shruthi.Thuravakkath@unisys.com

5th Bhavika Bhalgamiya
Enterprise Computing Solutions,Unisys
Pennsylvania, USA
Bhavika.Bhalgamiya@unisys.com

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***Abstract*—We propose an efficient application of the hybrid quantum-classical transfer learning model to improve the performance of credit risk classification. The Quantum Transfer Learning (QTL) model presented in this study consists of a frozen classical feature extractor and a quantum refinement layer which is trained on the features extracted from the real-world credit risk dataset. For baseline comparison, we evaluated a standalone Quantum Neural Network (QNN) that was trained directly on the credit risk dataset. Our experimental evaluations on a real-world normalized credit risk dataset reveal that the proposed QTL model significantly outperforms the standalone QNN and achieves results on par with the classical neural network.**

***Index Terms*—Quantum Machine Learning, Quantum Transfer Learning, Feedforward Neural Network, Quantum Neural Network, Varational Quantum Circuit**

I. Introduction

Given the growing demand for accurate and scalable credit risk assessment, it becomes important to explore frameworks like Quantum Transfer Learning (QTL) that can improve prediction quality while remaining computationally feasible under the limitations of current quantum hardware [1]. Furthermore, modern financial risk management increasingly incorporates machine learning and, more recently, quantum-enhanced models that aim to capture complex data dependencies and improve the prediction of minority classes. Transfer learning refers to the process of utilizing a model that has been pre-trained on a specific task and adapting it either wholly or partially to solve a related but distinct task. Beyond improving model generalization in low-data regimes, transfer learning contributes significantly to computational efficiency by reducing the need to train models from scratch. This leads to a reduction in overall training time and associated carbon emissions, making it an environmentally conscious approach in modern AI pipelines.

This study addresses the following core research challenges and contributions:

- **Integration of Quantum Transfer Learning (QTL) into Credit Risk Assessment:** Demonstrates how hybrid quantum-classical models can be effectively applied to real-world financial datasets to enhance classification.
- **Comprehensive Model Benchmarking:** Provides a detailed comparison of standalone Quantum Neural Network (QNN) and Hybrid QTL model, thereby offering a holistic perspective on quantum-assisted learning for financial prediction.
- **Improved Minority Class Detection:** Significantly enhances recall for the minority (defaulter) class by 0.62, addressing one of the most critical limitations of standalone QNN with recall 0.16.
- **Quantitative Performance Gains:** Reports improved area under the ROC curve (AUC up to 0.9108) and overall accuracy in QTL model.

II. Quantum Transfer Learning Architecture and Experimental set up

The hybrid Quantum Transfer Learning (QTL) architecture Fig. 1 is composed of a classical Feed Forward Neural Network and a QNN, arranged in a pipeline where the classical network serves as a frozen feature extractor.

Mathematical Formulation : We start with the input feature vector that is denoted as:

$$\mathbf{x} = [x_1, x_2, \ldots, x_n] \in \mathbb{R}^n \tag{1}$$

Here, n is the number of input features (e.g., $n = 23$ for the credit risk dataset). The classical feature extractor is a feedforward neural network that is represented as the following function:

$$\mathbf{z} = f_{\text{classical}}(\mathbf{x}; \theta_c) \tag{2}$$

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Here, θ_c represents the pre-trained parameters of the classical network, and $\mathbf{z} \in \mathbb{R}^d$ is the low-dimensional feature vector extracted from the final hidden layer (with $d = 4$). In Fig. 1 the classical network parameters θ_c are fixed and are not updated during training. These features $\mathbf{z} = [z_1, z_2, z_3, z_4]$ are then passed to the quantum circuit. Each component $z_i \in [-1, 1]$ is encoded into a single qubit using angle encoding via a parameterized rotational gate, typically the R_Y gate:

$$|\psi_i\rangle = \mathrm{R}_Y(z_i)\,|0\rangle = \cos\left(\frac{z_i}{2}\right)|0\rangle + \sin\left(\frac{z_i}{2}\right)|1\rangle \tag{3}$$

$$|\psi\rangle = \mathrm{R}_Y(z_1) \otimes \mathrm{R}_Y(z_2) \otimes \mathrm{R}_Y(z_3) \otimes \mathrm{R}_Y(z_4)\,|0000\rangle \tag{4}$$

Eq. 4 presents the full quantum state of the 4 qubits system after encoding. This quantum state is then processed by a Variational Quantum Circuit (VQC) composed of Hadamard and trainable parameterized unitary $U(\theta_q)$ gates:

$$|\phi\rangle = U(\theta_q)\,|\psi\rangle \tag{5}$$

Here, θ_q are the parameters of the quantum circuit. For this study, we choose **only the quantum parameters θ_q to be optimized during training**, while the classical model remains frozen. As the last step, measurement is performed in the computational basis, and the output is used for classification [3]:

$$\hat{y} = \sigma\left(\mathbf{w}^\top \langle\phi|Z|\phi\rangle + b\right) \tag{6}$$

Here, $\sigma(\cdot)$ is the sigmoid activation function, and $\mathbf{w}, b$ are the weights and bias of the final classical output layer that maps the quantum circuit output to a probability score.

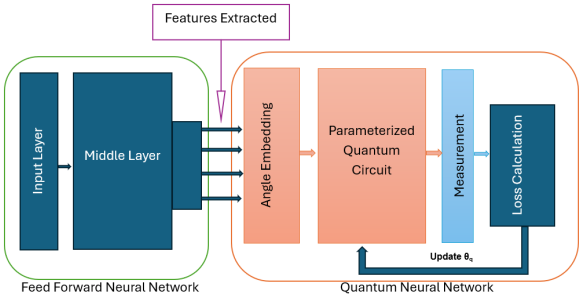

Fig. 1: Feature Extraction-Based QTL

The architecture consists of two primary stages: a pretrained classical feature extractor and a trainable VQC constructed using the PennyLane framework. The proposed QTL structure comprises of the following;

- **Dataset:** Kaggle Credit risk dataset [2] with normalized features and binary labels (loan default prediction) which contains 24 features and 32,580 data instances.
- **Split:** 80% training, 20% testing with stratification
- **Feature Extractor:** 3-layer Multi Layer Perceptron (MLP) with ReLU and Tanh activations
- **Quantum Circuit:** 4-qubit VQC (PennyLane, angle encoding + entanglement)
- **Loss Function:** Binary Cross Entropy
- **Optimizer:** Adam

III. Results and Conclusions

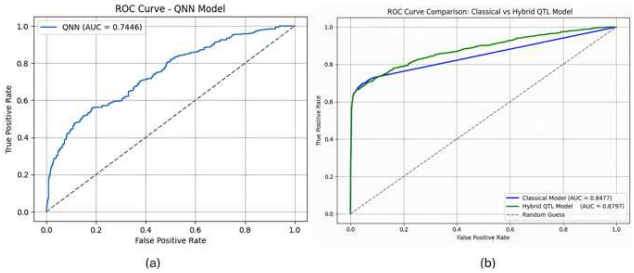

Fig. 2: ROC curve comparison

As observed in Fig.2 (b), Hybrid QTL model exhibits high accuracy (91.08%) and an AUC of 0.8797. It achieves a significant recall improvement for defaulters compared to the standalone QNN Fig. 2 (a) and outperforms the classical NN as also mentioned in Table. I. These results highlight our model's potential for real-world credit risk assessment under imbalanced data conditions.

TABLE I: Performance Comparison of Classical, QNN, and Hybrid QTL Models on Credit Risk Dataset

Metric	Classical Model	QNN	Hybrid QTL Model
Accuracy	0.9090	0.8097	0.9108
AUC Score	0.8477	0.7446	0.8797
Precision (Class 0)	0.90	0.81	0.90
Recall (Class 0)	0.99	0.99	0.99
F1-Score (Class 0)	0.94	0.89	0.95
Precision (Class 1)	0.95	0.82	0.95
Recall (Class 1)	0.61	0.16	0.62
F1-Score (Class 1)	0.75	0.27	0.75

We also demonstrated while the classical model performs well on the majority class, it lacks sensitivity to minority class defaulters. Although, the standalone QNN conceptually appealing, it suffers from poor recall, convergence inefficiencies and high training cost. This is because of its high computational demands, limited capacity to capture complex feature hierarchies without prior pre-processing, sensitivity to initialization, and gradient noise introduced by the parameter shift rule on simulated quantum circuits. In contrast, Hybrid QTL Model show marked improvements in recall and AUC. These findings underscore that hybrid architectures offer a meaningful path forward, balancing performance gains with the limitations of current quantum technologies.

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